

Exercises: Convex analysis

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Convex sets

Exercise 1 (Perspective function). Let $P : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ be the perspective function defined as $P(x, t) = x/t$, with $\text{dom}(P) = \mathbb{R}^n \times \mathbb{R}_+^*$.

1. Show that the image by P of the segment $[(\frac{x}{s}), (\frac{y}{t})]$ is the segment $[P((\frac{x}{s})), P((\frac{y}{t}))]$, i.e. $P([(x/s), (y/t)]) = [P((x/s)), P((y/t))]$.
2. Show that, if $C \subset \mathbb{R}^n \times \mathbb{R}_+^*$ is convex, then $P(C)$ is convex.
3. Show that, if $D \subset \mathbb{R}^n$, then $P^{-1}(D)$ is convex.

Exercise 2 (Dual cones). Recall that, for any set $K \subset \mathbb{R}^n$, $K^\oplus := \{y \in \mathbb{R}^n \mid \forall x \in K, \langle y, x \rangle \geq 0\}$. We say that K is self dual (which means that K is a closed convex cone) if $K^\oplus = K$.

1. Show that $K = \mathbb{R}_+^n$ is self dual.
2. We consider the set of symmetric matrices S_n with the scalar product $\langle A, B \rangle = \text{tr}(AB)$. Show that $K = S_n^+(\mathbb{R})$ is self dual.
3. Let $\|\cdot\|$ be a norm, show that $K = \{(x, t) \mid \|x\| \leq t\}$ has for dual $K^\oplus = \{(z, \lambda) \mid \|z\|_* \leq \lambda\}$, where $\|z\|_* := \sup_{x: \|x\| \leq 1} z^\top x$.

Exercise 3 (Normal cones of standard convex sets). Compute the normal cone $N_C(x)$ for the following closed convex sets:

1. (Euclidean ball) $C = \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$.
2. (Simplex) $\Delta = \{x \in \mathbb{R}^n : x \geq 0, \mathbf{1}^\top x = 1\}$.

Bonus. For the ball, compute the tangent cone $T_C(x)$ and verify $[T_C(x)]^\oplus = -N_C(x)$.

Convex functions

Exercise 4 (Recognizing convexity / strict / strong). For each function below, give: (i) its domain, (ii) whether it is convex, strictly convex, strongly convex on its domain (and provide a short justification).

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $\lambda > 0$, and assume A has full column rank when stated.

1. $f_1(x) = \|Ax - b\|_2$.
2. $f_2(x) = \frac{1}{2}\|Ax - b\|_2^2$.
3. $f_3(x) = \frac{1}{2}\|x\|_2^2 + \lambda\|x\|_1$.
4. $f_4(x) = -\log(b - a^\top x)$ with $\text{dom}(f_4) = \{x : a^\top x < b\}$.

Bonus. For f_2 , give a strong convexity modulus in terms of A (when A has full column rank).

Exercise 5 (Moving average). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function.

1. Show that, $s \mapsto \int_0^1 f(st)dt$ is convex.
2. Show that, $\mathbb{R}_+^* \ni T \mapsto 1/T \int_0^T f(t)dt$ is convex.

Exercise 6 (Partial infimum). Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ be a convex function and $C \subset \mathbb{R}^m$ a convex set. Show that the function

$$g : x \mapsto \inf_{y \in C} f(x, y)$$

is convex.

Exercise 7 (log determinant). Let, for any $X \in S_n$, $f(X) = \ln(\det(X))$ for $X \succ 0$, $-\infty$ otherwise. Consider, for $Z \succ 0$, and $V \in S_n$, the function $g : t \mapsto f(Z + tV)$.

1. Show that $g(t) = \sum_{i=1}^n \ln(1 + t\lambda_i) + f(Z)$, where the λ_i are the eigenvalues of $Z^{-1/2}VZ^{-1/2}$.

2. Show that g is concave. Conclude that f is concave.

Exercise 8 (Perspective function). Let $\phi : E \rightarrow \mathbb{R} \cup \{+\infty\}$. The perspective of ϕ is defined as $\tilde{\phi} : \mathbb{R}_+^* \times E \rightarrow \mathbb{R}$ by

$$\tilde{\phi}(\eta, y) := \eta\phi(y/\eta).$$

Show that ϕ is convex iff $\tilde{\phi}$ is convex.

Fenchel transform and subdifferential

Exercise 9 (Norm). Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$ and $\|y\|_* := \sup_{x: \|x\| \leq 1} y^\top x$ be its dual norm. Let $f : x \mapsto \|x\|$. Compute f^* and $\partial f(0)$.

Exercise 10 (Lasso). Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $\lambda > 0$ and consider

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1.$$

1. Show that the problem admits at least one solution, and is unique if A has full column rank.

2. Compute $\partial \|x\|_1$ and derive the optimality condition for a minimizer $x^\sharp$:

$$0 \in A^\top (Ax^\sharp - b) + \lambda \partial \|x^\sharp\|_1.$$

3. Prove the coordinate-wise characterization:

$$x_i^\sharp \neq 0 \Rightarrow a_i^\top (Ax^\sharp - b) = -\lambda \text{sign}(x_i^\sharp)$$

$$x_i^\sharp = 0 \Rightarrow |a_i^\top (Ax^\sharp - b)| \leq \lambda,$$

where a_i is the i -th column of A .

4. If $\lambda \geq \|A^\top b\|_\infty$, then $x^\sharp = 0$ is optimal.

5. Interpretation: explain in one sentence why large λ promotes sparsity of $x^\sharp$.

Exercise 11 (Fenchel calculus: indicator, support, and affine change). Let $C \subset \mathbb{R}^n$ be a nonempty closed convex set and define the indicator $\mathbb{I}_C$ and the support function

$$\sigma_C(y) := \sup_{x \in C} y^\top x.$$

1. Show that $(\mathbb{I}_C)^* = \sigma_C$.

2. Compute σ_C for:

$$(a) C = B_2(0, 1) = \{x : \|x\|_2 \leq 1\},$$

$$(b) C = \{x : \|x\|_\infty \leq 1\}.$$

3. Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and define $f(x) = \mathbb{I}_C(Ax + b)$. Give an expression for f^* (you may state the formula with the condition under which it holds).

Exercise 12 (Log sum exp). We consider $f(x) := \ln(\sum_{i=1}^n e^{x_i})$.

1. Show that f is convex. Hint : recall Holder's inequality $x^\top y \leq \|x\|_p \|y\|_q$ for $1/p + 1/q = 1$.

2. Show that $f^*(y) = \sum_{i=1}^n y_i \ln(y_i)$ if $y \geq 0$ and $\sum_i y_i = 1$, $+\infty$ otherwise.